

# A Lowrate Variable-Bias Integrate-and-Fire Time Encoding Machine

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**Abstract**—Integrate-and-fire time-encoding machines (IF-TEMs) are an alternative to conventional uniform sampling, where the signals are measured based on their variations. While IF-TEMs excel in energy efficiency through their event-driven sampling approach, they often oversample significantly to ensure accurate signal reconstruction. The adaptive IF-TEM method proposed in the literature addresses oversampling but lacks theoretical guarantees. In this work, we proposed a variable-bias IF-TEM that uses signals' bandwidth and energy to reduce oversampling. We derived theoretical guarantees and ensured that the method always results in perfect reconstruction while reducing oversampling. We presented simulation results to support the claims and show that the proposed method results in lower error and fewer samples than the existing techniques.

**Index Terms**—Integrate-and-fire time encoding machines, asynchronous sampling, lowrate sampling, variable bias, adaptive IF-TEM.

## I. INTRODUCTION

Sampling is a fundamental process in signal processing that represents continuous-time signals discretely, enabling efficient handling of analog signals using digital signal processors [1], [2]. A widely adopted method is the Shannon-Nyquist sampling theory, where analog signals are sampled uniformly using a global clock and reconstructed perfectly while sampling above the Nyquist rate [3].

An alternative approach, integrate-and-fire time-encoding machines (IF-TEM), offers asynchronous digital representation without the need for a global clock and with reduced power consumption [4]–[19]. In IF-TEM, analog signals are encoded based on the times when the signal or its function crosses a set threshold, with the rate of these crossings or firing rate (FR) proportional to the local content of the signal. For instance, rapid signal variations result in more frequent firings compared to slower variations. Specifically, in an IF-TEM, the analog signal undergoes biasing to ensure positivity, followed by scaling, integration, and comparison against a threshold. Whenever the threshold is reached, firing instants are recorded, thereby encoding the analog signal information [5], [20].

Recent advancements in time-based sampling theories have shown TEM's capability to sample and reconstruct bandlimited signals accurately [6], [17], [19]–[23]. The FR of an IF-TEM is lower and upper-bounded, and the bounds are functions of the TEM parameters and the maximum amplitude of the signal. The minimum firing rate required for perfect recovery of a

bandlimited signal should be greater than its Nyquist rate. This condition leads to undesirable oversampling.

Recently, Omar and Cohen [24] suggested an adaptive IF-TEM (AIF-TEM) where the bias varies based on the signal's amplitude in a time interval. To be precise, the authors suggested updating bias values after each firing. The update should be a function of the expected maximum value of the signal after each firing instant. Note that accurately estimating the maximum amplitude for the signal and then the bias for the subsequent firing is key in the AIF-TEM. The variable bias results in local bounds on the FRs, unlike a global bound in a conventional IF-TEM. These local bounds, especially the upper bound, which is the reason for oversampling, are typically much smaller than the global upper bound. Consequently, the average oversampling could be much smaller than the one in the conventional IF-TEM. In addition, the authors derived bounds on quantization error for the adaptive IF-TEM.

The AIF-TEM introduced in [24] presents a new approach to tackle the critical oversampling issue. However, the approach has certain limitations that may constrain its practical application. A primary concern is the absence of a well-defined method for estimating bias using given firing instants. In their simulations, the authors suggest a recursive approach to predict peak signal values and biases for upcoming firings, yet they lack theoretical justification. Specifically, it was not shown that this recursive method consistently maintains a feasible bias, that is, ensuring a positively biased signal. Additionally, how the update rule should adjust for the change in the signal's bandwidth or any other signal parameters was not discussed. For the decoding process, the authors propose a segment-wise reconstruction where bias updates rely on a fixed number of past firings. However, the choice of segment size can significantly impact reconstruction accuracy.

In our work, we explore an IF-TEM-based discretization approach with variable bias. Our variable-bias IF-TEM (VB-IF-TEM) method adjusts bias following each IF-TEM firing to ensure the signal remains positive until the next firing. Leveraging properties of bandlimited signals such as bandwidth and maximum energy, we establish bounds for the signal's next firing. Unlike the bias update rule in [24], we theoretically demonstrate that our approach consistently maintains a feasible bias, that is, the biased signal is non-negative. Further, we derived bounds on the oversampling factor. We show that our proposed bias update method achieves perfect reconstruction for bandlimited signals with an upper bound on oversampling

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rate of 4.23 times the Nyquist rate. Simulation results show that the proposed method results in lower reconstruction errors than the existing methods. This approach promises improved power efficiency through reduced oversampling.

In the following section, we formulate the problem. In Section III, we present the proposed VB-IF-TEM. We conclude in Section IV.

## II. PROBLEM FORMULATION

Consider a class of finite-energy bandlimited signals  $\mathcal{B}_{\Omega_0, E}$  such that if  $f(t) \in \mathcal{B}_{\Omega_0, E}$  then the Fourier transform of  $f(t)$  vanishes outside the interval  $[-\Omega_0, \Omega_0]$  and  $\|f(t)\|_2^2 \leq E$ . In addition, the signals in this class are always bounded. Precisely, in [25] it was shown that

$$|f(t)| \leq c = \sqrt{E \Omega_0 / \pi}. \quad (1)$$

In the conventional uniform sampling, signals in  $\mathcal{B}_{\Omega_0, E}$  sampled at least the Nyquist rate  $2\Omega_0$  can then be reconstructed perfectly. Specifically, for any  $f(t) \in \mathcal{B}_{\Omega_0, E}$ , the uniform samples  $f(nT_s), n \in \mathbb{Z}$  are sufficient for recovery provided that  $T_s \leq \frac{\pi}{\Omega_0}$ . The Shannon-Nyquist interpolation is invoked for perfect reconstruction:

$$f(t) = \sum_{n=-\infty}^{\infty} f(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right) \quad (2)$$

where  $\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$ . In this sampling scheme, the sampling rate is a function of the bandwidth, but it is independent of the signal's dynamic range. Further, the sampling interval  $T_s$  is fixed, and only the amplitudes  $f(nT_s)$  are measured.

In the IF-TEM-based sampling scheme, a signal  $f(t)$  is represented by a set of time instants  $\{t_n\}_{n \in \mathbb{Z}}$  that are measured as follows. The signal is added with a bias  $b(t)$  such that  $f(t) + b(t) > 0$ . The signal is then divided by  $\kappa$  and integrated, as shown in Fig. 1. The integral is then compared to a threshold  $\Delta$ . The time instants at which the output is equal to the threshold are recorded as the firing instants  $\{t_n\}_{n \in \mathbb{Z}}$ . After each firing, the integration is reset. Mathematically, the firing instants, the IF-TEM parameters  $\kappa, \Delta$ , and the bias  $b(t)$  are related to the signal  $f(t)$  as

$$\frac{1}{\kappa} \int_{t_{n-1}}^{t_n} f(t) + b(t) dt = \Delta. \quad (3)$$

In conventional IF-TEMs, the bias  $b(t)$  is kept constant as  $b(t) = b$ , where  $b > c$ . With constant bias, the IF-TEM relationship in (3) is given as

$$\int_{t_{n-1}}^{t_n} f(t) dt = \kappa \Delta - b(t_n - t_{n-1}). \quad (4)$$

From (4), we have the following inequalities [5]

$$\frac{\kappa \Delta}{b - c} \leq t_n - t_{n-1} \leq \frac{\kappa \Delta}{b + c}. \quad (5)$$

By using the inequalities, we have the maximum and minimum firing rates, that is, the maximum and minimum number of time encodings per second, respectively, given as

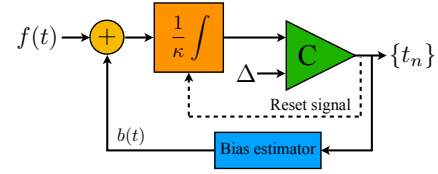


Fig. 1. Schematic of Variable-bias IF-TEM

$$\text{FR}_{\max} = \frac{b + c}{\kappa \Delta} \quad \text{and} \quad \text{FR}_{\min} = \frac{b - c}{\kappa \Delta}. \quad (6)$$

The FR-bounds play a key role in reconstructing the signals from their time encodings and the oversampling problem considered in this work. The reconstruction process, that is, to determine  $f(t)$  from  $\{t_n\}_{n \in \mathbb{Z}}$ , an iterative method is typically applied [5]. It was shown that the algorithm results in perfect reconstruction if

$$\text{FR}_{\min} = \frac{b - c}{\kappa \Delta} > \frac{\Omega_0}{\pi}. \quad (7)$$

In other words, the minimum firing rate should be greater than the Nyquist rate. This also implies that the difference between two successive firing instants,  $t_n - t_{n-1}$ , can not exceed the Nyquist interval  $T_{Nyq} = \frac{\pi}{\Omega_0}$ . For the signals in  $\mathcal{B}_{\Omega_0, E}$ , the condition in (7) can be fulfilled by choosing  $b, \kappa$ , and  $\Delta$  carefully.

Though perfect recovery is always achieved with (7), an IF-TEM always operates in the oversampling region where the number of measurements is greater than the Nyquist rate. For example, for a fixed  $\kappa$  and  $\Delta$ , consider the following choice of bias  $b = \gamma \frac{\Omega_0}{\pi} \kappa \Delta + c$ , where  $\gamma > 1$ . The choice of  $b$  ensures that (7) is satisfied. In this setup,  $\text{FR}_{\max}$  is given as  $\text{FR}_{\max} = \gamma \frac{\Omega_0}{\pi} + \frac{2c}{\kappa \Delta}$ . The oversampling rate, is proportional to the signal's peak amplitude and could be significantly higher than the Nyquist rate.

Next, we discuss the proposed solution with a variable bias such that the oversampling can be within a limit and signal-based measurements are achieved.

## III. A VARIABLE BIAS IF-TEM

In this section, we discuss the proposed VB-IF-TEM that will address the oversampling issue. To motivate, we depict an example in Fig. 2, where firings of a conventional IF-TEM and a variable-bias IF-TEM are shown. The bias is changed based on the signal amplitude in the latter case. We observe a considerable reduction in the firings, especially in low-amplitude regions. In this case, we assume the knowledge of the signal's local amplitude to set the bias, which is unknown in practice. In the following, we discuss our bias estimation approach that only uses the signal's bandwidth and energy.

### A. Proposed Bias Estimation

Consider a signal  $f(t) \in \mathcal{B}_{\Omega_0, E}$ , which is time-encoded by using the proposed VB-IF-TEM, where the bias is updated after each firing. Let the current firing instant be  $t_n$ .

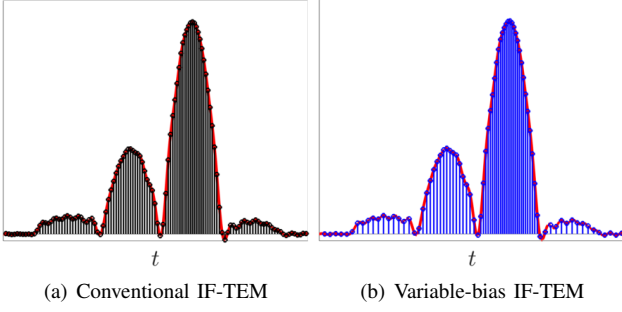


Fig. 2. A comparison of a conventional IF-TEM and a VB-IF-TEM: The VB-IF-TEM results in 80 firings, compared to 118 in the conventional one.

We assume that the bias added to  $f(t)$  after the firing at  $t_{n-1}$  is  $b_{n-1}$  and its negative is lesser than or equal to  $c_{n-1} = \min_{t \in (t_{n-1}, t_n]} f(t)$ . The bias is feasible and ensures that  $f(t) + b_{n-1} > 0$  for  $t \in (t_{n-1}, t_n]$ . Next, we discuss an approach to estimate  $c_n$  such that  $f(t) \geq c_n$  from  $t_n$  to the next firing instant  $t_{n+1}$ . Then choosing a bias  $b_n \geq c_n$  ensures that  $f(t) + b_n$  is non-negative from  $t_n$  to the next firing.

Simply put, here is our approach: To estimate bias within the interval  $[t_n, t_{n+1}]$ , we begin by determining the potential range of values for  $t_{n+1}$ . Next, we assess the minimum value the signal can reach during this interval. To achieve this, we initially establish the range of  $f(t_n)$  and apply it to solve the estimation issues mentioned.

Our bias estimation approach uses the current firings and bias values together with the rate of change of the bandlimited signals in  $\mathcal{B}_{\Omega_0, E}$  given as [25]

$$\left| \frac{df(t)}{dt} \right| < \Omega \sqrt{\frac{E\Omega}{3\pi}} = \epsilon. \quad (8)$$

To begin with, consider the following signal average computed from the time-encodings  $t_{n-1}, t_n$ , and the bias  $b_{n-1}$ :

$$\hat{f}_n = \frac{1}{t_n - t_{n-1}} \int_{t_{n-1}}^{t_n} f(t) dt = \frac{\kappa \Delta}{t_n - t_{n-1}} - b_{n-1}. \quad (9)$$

Being bandlimited with finite energy,  $f(t)$  is continuous, and hence, by mean value theorem, there exists at least one  $\xi_n \in (t_{n-1}, t_n)$ , such that  $f(\xi_n) = \hat{f}_n$ . By using the average value and (8), we first find a lower bound on  $f(t_n)$ , which will be subsequently used to estimate  $c_n$ .

From the time instant  $\xi_n$ , the value of  $f(t)$  can increase (decrease) with the maximum (minimum) slope  $\epsilon$  as shown in Fig. 3. The intersections of straight lines with slopes  $\pm\epsilon$ , and value  $\hat{f}_n$  at  $t = \xi_n$ , with a vertical line at  $t_n$  determines the range of  $f(t_n)$ . Mathematically, by using (8), we have that

$$f(\xi_n) - (t_n - \xi_n)\epsilon \leq f(t_n) \leq f(\xi_n) + (t_n - \xi_n)\epsilon. \quad (10)$$

The lower bound, which of the interest, is a function of  $\xi_n$ . To achieve the lowest bound, the signal has to follow the straight line with slope  $-\epsilon$  for  $t \in [\xi_n, t_n]$  and, additionally, satisfy the constraint  $\int_{t_{n-1}}^{\xi_n} (f(t) - \hat{f}_n) dt = \int_{\xi_n}^{t_n} (\hat{f}_n - f(t)) dt$ . The equality constraint is a consequence of the fact that  $\int_{t_{n-1}}^{t_n} (f(t) - \hat{f}_n) dt = 0$ . It can be shown that the bound

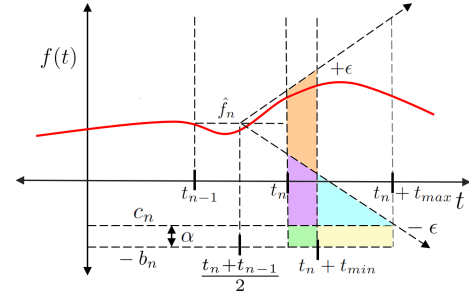


Fig. 3. Visualization of the proposed bias and firing rate estimation approach.

$f(\xi_n) - (t_n - \xi_n)\epsilon$  is lowest while satisfying the equality constraint for  $\xi_n = \frac{t_n + t_{n-1}}{2}$ . Using similar arguments to find the upper bound as well. Hence, we have that

$$\hat{f}_n - \frac{t_n - t_{n-1}}{2}\epsilon \leq f(t_n) \leq \hat{f}_n + \frac{t_n - t_{n-1}}{2}\epsilon. \quad (11)$$

Since the bias is added such that the signal remains positive, next, we focus on the minimum value the signal can achieve within the interval  $[t_n, t_{n+1}]$ . Though  $t_{n+1}$  is unknown, there always exist  $t_{\min}$  and  $t_{\max}$  such that

$$t_{\min} \leq t_{n+1} - t_n \leq t_{\max}. \quad (12)$$

For the following discussion, we assume that  $t_{\min}$  and  $t_{\max}$  are known and later discuss how to determine them. Nevertheless, we would like to mention that  $t_{n+1} = t_n + t_{\max}$  when the signal decreases at a rate  $-\epsilon$  from its value at  $t = t_n$ . On the other hand, when the signal rapidly increases from its value at  $t = t_n$ , the next firing happens quickly at  $t = t_n + t_{\min}$ . Next, we determine the lower bound on the signal at  $t = t_n + t_{\max}$ , which would aid the bias computation.

Starting from the value  $f(t_n)$  at  $t = t_n$  and using (8), the signal  $f(t)$  at  $t = t_n + t_{\max}$  is bounded as

$$f(t_n) - t_{\max}\epsilon \leq f(t_n + t_{\max}). \quad (13)$$

By using the lower bounds in (11), we have that

$$f(t_n + t_{\max}) \geq \hat{f}_n - \frac{(t_n - t_{n-1})}{2}\epsilon - t_{\max}\epsilon = c_n. \quad (14)$$

Then, choosing a bias as

$$b_n + c_n = \alpha, \quad (15)$$

where  $\alpha \geq 0$ , ensures that  $f(t) + b_n \geq 0$  for  $t \in [t_n, t_n + t_{\max}]$ . Our bias estimation solely depends on the previous two firing instants, previous bias value,  $\alpha$ , and  $t_{\max}$  for the class of bandlimited signals.

Having established that the proposed approach results in a feasible bias for successive firings, we next discuss the computation of  $t_{\max}$  and  $t_{\min}$ , then discuss conditions for perfect reconstruction.

### B. Bounds on Firing Rates

The firing  $t_{n+1}$  will occur at  $t_n + t_{\max}$  if the signal decays at the maximum possible rate  $\epsilon$ . In this case, the area of the

shaded region, except for the region in orange, in Fig. 3 is equal to  $\kappa\Delta$ , that is,

$$\int_{t=t_n}^{t_n+t_{\max}} (f(t) + b_n)dt = \alpha t_{\max} + \frac{1}{2}\epsilon t_{\max}^2 = \kappa\Delta \quad (16)$$

Hence, we have that

$$t_{\max} = \frac{-\alpha + \sqrt{\alpha^2 + 2\epsilon\kappa\Delta}}{\epsilon}. \quad (17)$$

Next, we determine  $t_{\min}$ . Note that  $t_n = t_{n-1} + t_{\min}$  when signal value increases at a rate  $\epsilon$ . Then, as in (13), the upper bound is given as

$$f(t_n + t_{\min}) \leq f(t_n) + t_{\min}\epsilon. \quad (18)$$

Then by using the upper bound on  $f(t_n)$  (cf. (11)), the areas of the signal between  $t_n$  to  $t_n + t_{\min}$ , indicated by green, purple and orange regions as shown in Fig. 2, is equal to  $\Delta$ . Mathematically, we have

$$\begin{aligned} \Delta &= \alpha t_{\min} + \frac{t_{\min}}{2} (\epsilon t_{\max} + \epsilon(t_{\max} - t_{\min})) \\ &+ \frac{t_{\min}}{2} (\epsilon(t_n - t_{n-1}) + \epsilon(t_n - t_{n-1} + 2t_{\min})). \end{aligned} \quad (19)$$

For  $\alpha = 0$  (lowest possible bias) and for the lowest value of  $f(t_n)$  given  $\hat{f}_n$ , we have  $t_n - t_{n-1} = t_{\max}$ , which leads to

$$t_{\min} = \frac{-2\epsilon t_{\max} + \sqrt{\epsilon^2(2t_{\max})^2 + 2\kappa\Delta\epsilon}}{\epsilon} \quad (20)$$

Solving this, we obtain,  $t_{\min} = (\sqrt{5} - 2)\sqrt{\frac{2\kappa\Delta}{\epsilon}}$  and  $t_{\max} = \sqrt{\frac{2\kappa\Delta}{\epsilon}}$ . Then from (12), we have the following bounds on the firing rates for variable bias:

$$\text{FR}_{\min}^{\text{vb}} = \sqrt{\frac{\epsilon}{2\kappa\Delta}} \quad \text{and} \quad \text{FR}_{\max}^{\text{vb}} = (\sqrt{5} + 2)\sqrt{\frac{\epsilon}{2\kappa\Delta}}. \quad (21)$$

Next, we consider perfect reconstruction in these settings.

### C. Perfect Reconstruction and Oversampling

For reconstructing the signal from the firings, we can employ the iterative algorithm proposed in [5] that uses the average values  $\int_{t_{n-1}}^{t_n} f(t)dt = \kappa\Delta + b_{n-1}$ . For perfect reconstruction, it is required that

$$\text{FR}_{\min}^{\text{vb}} = \sqrt{\frac{\epsilon}{2\kappa\Delta}} < \frac{\Omega_0}{\pi}. \quad (22)$$

As  $\epsilon$  is fixed for the class of signals, in order to satisfy the above equation, we must choose the values of  $\kappa$  and  $\Delta$  accordingly.

By setting  $\text{FR}_{\min}^{\text{vb}} = \gamma \frac{\Omega_0}{\pi}$ , we have  $\text{FR}_{\max}^{\text{vb}} = (\sqrt{5} + 2)\gamma \frac{\Omega_0}{\pi}$ , where  $\gamma > 1$ . As  $\gamma \rightarrow 1$ , we achieve a tighter bound on the oversampling factor, which is upper bounded by  $\sqrt{5} + 2$  or 4.23.

### D. Results and Comparisons

Simulations are presented here to compare VB-IF-TEM reconstruction to the conventional IF-TEM, and the AIF-

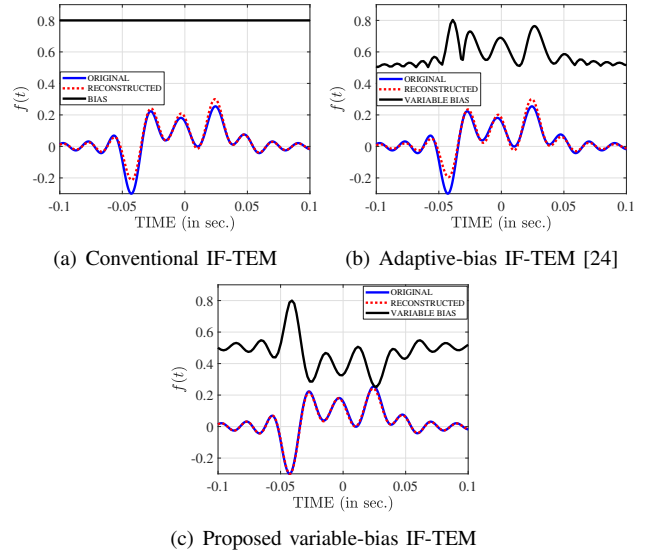


Fig. 4. A comparison of different IF-TEMs: (a) conventional IF-TEM - MSE = -38.99 dB with 726 firings; (b) AIF-TEM - MSE = -37.69 dB with 478 firings; and (c) proposed VB-IF-TEM - MSE = -50.97 dB with 454 firings.

TEM [24]. We generated an input signal as  $x(t) = \sum_{n=-M}^M a[n] \frac{\sin(\Omega_0 t - n\pi)}{\Omega_0 t - n\pi}$ , where  $\Omega_0 = 2\pi 50$  rad./sec, and  $a[n]$ s were randomly generated to have  $c = 0.3$  and  $E = 0.0023376$ . We set  $\kappa = 1$  and  $\Delta = 0.001$  for all the IF-TEMs. The bias was set as  $b = 0.8$  for the conventional method. For AIF-TEM, the parameters were set as  $\alpha_1 = 0.98$ ,  $\alpha_2 = 0.17$ ,  $\beta = 0.5$ ,  $b_{\min} = 0.3$ , and  $w = 1$ . The choices ensure that the reconstruction error is as low as possible with the least firings for the signal.

The true signal, reconstructed ones, and the biases are shown in Fig. 4. We note that the mean-squared error (MSE) for VB-IF-TEM is -50.97 dB, whereas the MSEs for the conventional and adaptive IF-TEMs are -37.69 dB and -38.99 dB, respectively. Further, the proposed and adaptive-bias methods [24] have 454 and 478 firings, respectively, compared to 726 in the conventional one. The results show that fewer firings could be used for better reconstruction with variable bias. We also plotted the bias values for the proposed and the AIF-TEM and noted that our approach is able to follow the signal closely while estimating the bias.

## IV. CONCLUSION

Our work introduces a VB-IF-TEM method that dynamically adjusts the bias after each firing to maintain signal positivity, leveraging the characteristics of bandlimited signals such as bandwidth and maximum energy. We provide theoretical guarantees for maintaining a feasible bias, differentiating our approach from previous methods. Additionally, we derive oversampling factor bounds and demonstrate that our method achieves near-perfect signal reconstruction with an upper bound on oversampling rate approaching 4.23 times the Nyquist rate, significantly improving power efficiency by reducing oversampling. This approach offers a promising solution for more efficient signal discretization.

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